

1. The figure shows the velocity profile of laminar flow in a circular pipe. The shape is a parabola so that at the wall of the pipe, $du/dr = -2U_{CL}/(D/2)$. Work out the “friction coefficient” $C_f = \tau_w / (\frac{1}{2} \rho U_{CL}^2)$ as a function of U_{CL} , D , ρ , μ and demonstrate that $C_f = 8/Re_D$ where Re_D is based on the centreline velocity U_{CL} . Evaluate C_f for airflow in a pipe of 8mm diameter with a centre-line speed of 3ms^{-1} assuming $\rho = 1.19\text{ kg m}^{-3}$, $\mu = 1.82 \times 10^{-5}\text{ kg m}^{-1}\text{s}^{-1}$. If the centre-line velocity is raised to 15ms^{-1} would you expect the flow to be laminar or turbulent? (Ans: $C_f = 5.1 \times 10^{-3}$)

Note: the friction coefficient is usually based on the average velocity over the cross section, which for a paraboloid profile in a circular pipe is $U_{CL}/2$.

$$\left. \frac{du}{dr} \right|_w = \frac{-2U_{CL}}{\frac{D}{2}} = \frac{-4U_{CL}}{D}, \quad U_{\infty} = U_{CL}$$

$$\tau_w = \mu \left. \frac{du}{dr} \right|_w = \mu \frac{4U_{CL}}{D}$$

$$\begin{aligned} \therefore C_f &= \frac{\tau_w}{\frac{1}{2} \rho (U_{\infty})^2} = \frac{\mu \frac{4U_{CL}}{D}}{\frac{1}{2} \rho (U_{CL})^2} = \frac{8\mu}{\rho U_{CL} D} \\ &= \frac{8}{Re_D} \end{aligned}$$

$$\text{When } U_{CL} = 3\text{ m/s}$$

$$\therefore C_f = \frac{8 \times 1.82 \times 10^{-5}\text{ kg/ms}}{1.19\text{ kg/m}^3 \times 3\text{ m/s} \times 8\text{ mm}} = 5.098 \times 10^{-3}$$

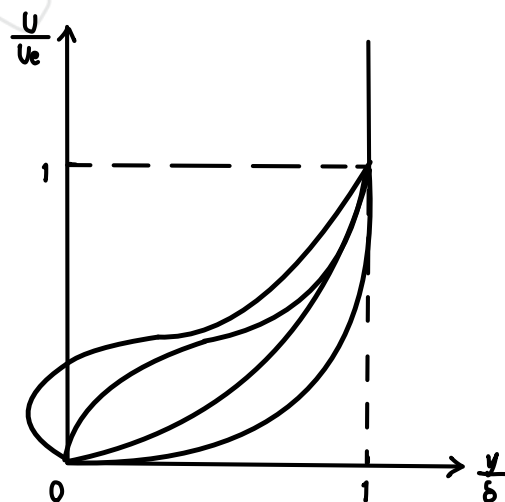
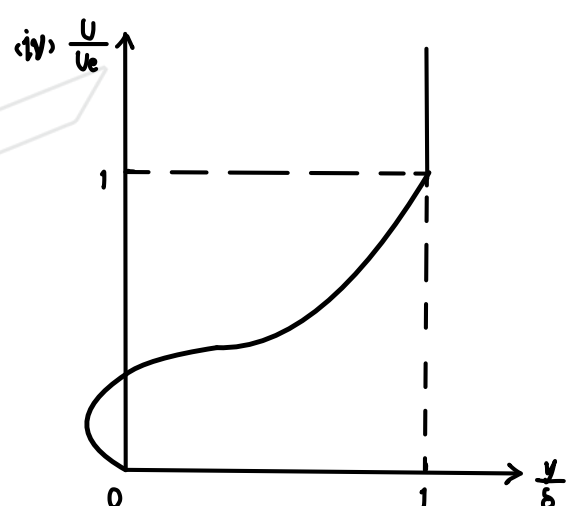
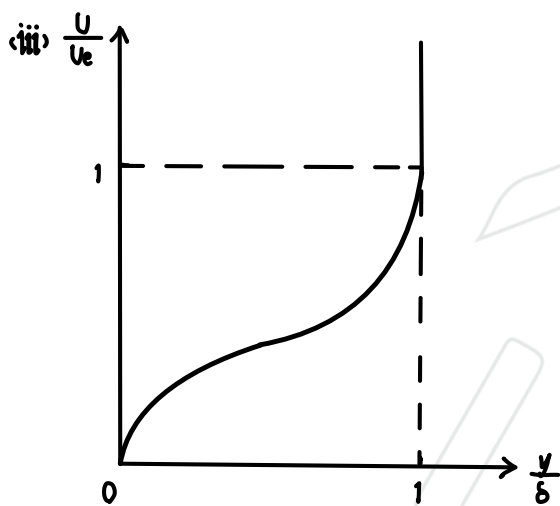
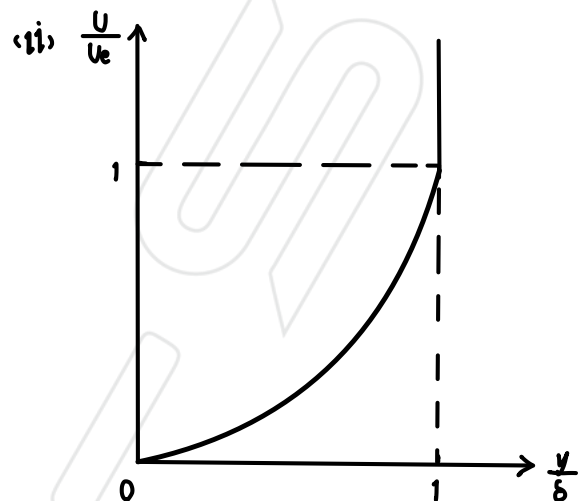
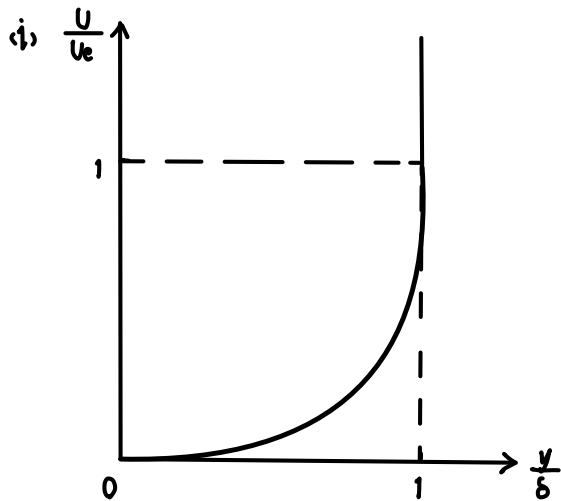
$$\text{When } U_{CL} = 15\text{ m/s}$$

$$\therefore C_{f2} = \frac{3\text{ m/s}}{15\text{ m/s}} C_f = 1.0196 \times 10^{-3}$$

$$\therefore Re_{D2} = \frac{8}{C_{f2}} = 7846 > 2500 \quad \therefore \text{The flow will be turbulent}$$

2. Draw velocity profiles, as u/u_e vs. y/δ for laminar boundary layers in

- (i) favourable pressure gradient
- (ii) zero pressure gradient
- (i) adverse pressure gradient
- (ii) separated flow



3. Calculate the thickness of the laminar boundary layer on a flat plate at distances of 1m and 2m from the leading edge, for free stream velocities of 2m/s and 4m/s. Compare the results and discuss. (Ans at 1m: 14.6mm, 10.4mm at 2m: 20.8mm, 14.6mm)

On a flat plate, $\frac{dp}{dx} = 0$

$$\therefore \delta = 5.3 \sqrt{\frac{\nu x}{U_e}}$$

For air in normal condition,

$$\nu = \frac{\mu}{\rho} = \frac{1.82 \times 10^{-5} \text{ kg/ms}}{1.19 \text{ kg/m}^3} = 1.52941 \times 10^{-5} \text{ m}^2/\text{s}$$

When $x = 1\text{m}$, $U_e = 2\text{m/s}$

$$\delta = 0.0147\text{m} = 14.7\text{mm}$$

When $x = 1\text{m}$, $U_e = 4\text{m/s}$

$$\delta = 0.0104\text{m} = 10.4\text{mm}$$

When $x = 2\text{m}$, $U_e = 2\text{m/s}$

$$\delta = 0.0207\text{m} = 20.7\text{mm}$$

When $x = 2\text{m}$, $U_e = 4\text{m/s}$

$$\delta = 0.0147\text{m} = 14.7\text{mm}$$

$\therefore \delta \uparrow$ when $x \uparrow$ while $\delta \downarrow$ when $U_e \uparrow$

4. The velocity U within the boundary layer on an aerofoil at a certain position from the leading edge and a distance y from a surface, is given by $\frac{U}{U_e} = 2\frac{y}{\delta} - \left(\frac{y}{\delta}\right)^2$. U_e is the velocity at the edge of the boundary layer and δ is the boundary layer thickness. If the aerofoil is travelling through air at standard atmospheric conditions calculate the wall shear stress for $U_e = 30\text{m/s}$ and $\delta = 6\text{mm}$ and calculate the local pressure gradient along the aerofoil surface (Ans 0.182 Nm^{-2} , -30.33Nm^{-3})

$$\frac{U}{U_e} = 2 \frac{y}{\delta} - \left(\frac{y}{\delta}\right)^2$$

$$\therefore \frac{U}{30} = \frac{1000}{3}y - \frac{250000}{9}y^2$$

$$\therefore U = 10000y - \frac{2500000}{3}y^2$$

$$\therefore \frac{du}{dy} = 10000 - \frac{5000000}{3}y$$

$$\begin{aligned}\therefore \tau_w &= \mu \frac{du}{dy} \Big|_{y=0} = 1.82 \times 10^{-5} \times 10000 \\ &= 0.182 \text{ (N/m}^2\text{)}\end{aligned}$$

$$\therefore \frac{\partial u}{\partial t} + u \frac{du}{dx} + v \frac{du}{dy} + \frac{1}{\rho} \frac{dp}{dx} = \frac{\mu}{\rho} \frac{\partial^2 u}{\partial y^2}$$

$$\therefore \frac{dp}{dx} = \mu \frac{d^2u}{dy^2}$$

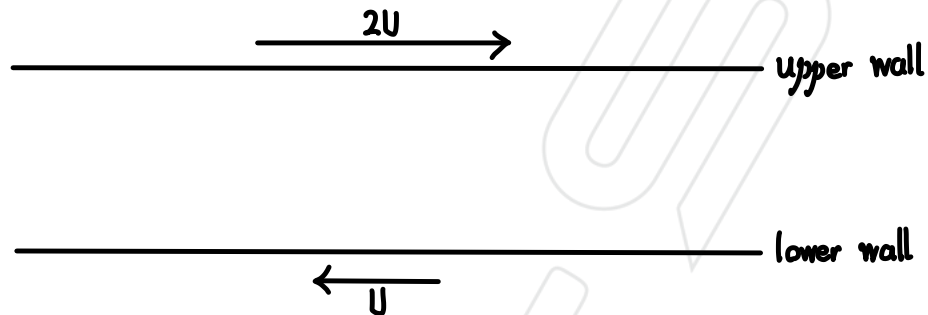
$$= 1.82 \times 10^{-5} \times \left(-\frac{5000000}{3}\right)$$

$$= -\frac{91}{3} \text{ (N/m}^3\text{)}$$

$$= -30.333 \text{ (N/m}^3\text{)}$$

5. A two-dimensional, parallel- side duct of height $2h$ has its upper wall moving with velocity $2U$ and its lower wall moving in the opposite direction with velocity U . The movement of the walls causes a flow in the duct. Starting from the equations of motion for steady viscous boundary layer flow, derive an expression for the velocity profile across the duct. Noting that $dp/dx = 0$ determine the distribution of velocity and vorticity across the duct.

(Ans $u(y) = U/2 + 3/2 Uy/h$, $\omega(y) = -3U/(2h)$ with $y=0$ in the middle of the duct)



$$\therefore \rho \frac{\partial u}{\partial x} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + \frac{dp}{dx} = \mu \frac{d^2 u}{dy^2}$$

$\therefore$ In steady viscous boundary flow

$$\frac{dp}{dx} = \mu \frac{d^2 u}{dy^2} \quad \therefore \frac{d^2 u}{dy^2} = \frac{1}{\mu} \frac{dp}{dx}$$

$$\therefore \frac{du}{dy} = \frac{1}{\mu} \frac{dp}{dx} y + C \quad \therefore u = \frac{1}{2\mu} \frac{dp}{dx} y^2 + Cy + D$$

$$\text{When } \frac{dp}{dx} = 0 \quad \therefore \frac{du}{dy} = C \quad \therefore u = Cy + D$$

$$\text{When } y = h, u = 2U \quad \therefore 2U = Ch + D$$

$$\text{When } y = -h, u = -U \quad \therefore -U = -Ch + D$$

$$\therefore C = \frac{3U}{2h} \quad \therefore D = \frac{1}{2}U$$

$$\therefore u(y) = \frac{3U}{2h} y + \frac{1}{2}U$$

$$\omega(y) = \frac{dv}{dx} - \frac{du}{dy} = -\frac{du}{dy} = -\frac{3U}{2h}$$